

Haridev M

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Summary

Diploma graduate in Computer Science and Engineering, currently pursuing B.Tech in Computer Science and Engineering. Enthusiastic Flutter developer with experience in embedded systems and integrating mobile apps with web platforms. Passionate about building practical projects, exploring innovative technologies, and continuously improving technical skills.

Professional Skills

- **Mobile Development:** Flutter, Dart
- **Embedded Systems:** Arduino (Uno R3, Mega), Serial Communication
- **Databases:** SQLite, MySQL, Firebase, Supabase
- **Programming:** Python, Java, C
- **Web Development:** HTML, CSS, JavaScript
- **Machine Learning:** scikit-learn, pandas, XGBoost, SVM, SHAP
- **Tools & Frameworks:** Git, GitHub, Flask, OpenWeather API
- **Languages:** Malayalam (native), English

Education

Rajiv Gandhi Institute of Technology, Kottayam B.Tech in Computer Science and Engineering	Expected 2027
MVGM Government Polytechnic College, Vennikulam, Pathanamthitta, Kerala Diploma in Computer Engineering CGPA: 8.57	2021 – 2024
NSS Higher Secondary School, Kottayam, Kerala Certificate of Higher Education: Humanities	2021
SRV Peethom, Kottayam, Kerala GCSEs (General)	2019

Experience

Intern, Indian Institute of Information Technology (IIIT) Kottayam	May 1, 2025 – July 31, 2025
• Developed predictive models for dementia detection using XGBoost and SVM, achieving an accuracy improvement of over 10% compared to baseline models. • Conducted comprehensive data preprocessing, feature engineering, and analysis on healthcare datasets (OASIS, Kaggle) containing 400+ patient records. • Applied SHAP for explainability, identifying top 7 predictors such as MMSE, SES, nWBV, Education, Depression Status, APOE-ε4, and Smoking Status. • Presented analytical insights and visualization reports to research mentors, improving interpretability and collaboration outcomes. • Technologies: Python, scikit-learn, pandas, SHAP.	

Flutter Intern, Spectrum Softtech Solution (P) LTD

May 8, 2023 – May 31, 2023

- Learned Flutter fundamentals, including widget tree structure, state management, and responsive UI design
- Implemented authentication features using Firebase for secure user login and registration
- Developed a To-Do List mobile application using Flutter and Dart, integrating local storage with SQLite and task prioritization
- Gained hands-on experience with building functional layouts, navigation, and interactive app components

Trainee Engineer, Networkz Systems

May 5, 2022 – May 25, 2022

- Gained hands-on experience in Python programming and embedded systems (Arduino Uno R3, Mega).
- Developed and demonstrated a working traffic light system prototype, improving understanding of hardware-software interaction.

Projects

Gender Differences in Dementia Prediction (IIIT Kottayam)

2025

- Built gender-specific dementia prediction models using XGBoost and SVM on OASIS-1 and Kaggle datasets with 95% accuracy.
- Conducted feature selection, hyperparameter tuning, and 5-fold cross-validation for optimized performance.
- Applied SHAP for model interpretability, identifying MMSE, SES, nWBV, Education, and Depression Status as critical indicators.
- Presented detailed findings and insights to mentors, demonstrating advanced data analysis and reporting skills.

Smart Environment Control System (Hackathon)

2025

- Developed a smart college environment control system integrating Arduino sensors and OpenWeather API for real-time monitoring.
- Built a Flask-based web UI visualizing temperature and light data with 5-second auto-refresh for live updates.
- Provided automated environmental suggestions, improving classroom comfort and energy efficiency.
- Led a 3-member team to complete design, testing, and deployment within 24-hour hackathon constraints.

Smart Door Lock System

2024

- Designed a Bluetooth-enabled smart door lock using Arduino Mega and HC-05, enhancing security and remote control.
- Developed a mobile app with Firebase authentication supporting real-time lock/unlock and password management.
- Integrated LCD feedback and EEPROM-based password storage, improving reliability and user experience.

Study Assistant Chatbot (Hackathon)

2024

- Built an AI chatbot for KTU syllabus assistance using Flutter and integrated with Ollama on a local server.
- Designed features for summarization and Q&A, completed within an 18-hour hackathon (TinkerHub).
- Demonstrated rapid prototyping, team collaboration, and applied AI integration in educational tools.

To-Do List App (Flutter)

2023

- Developed a cross-platform task management app using Flutter and Dart, improving daily productivity.
- Integrated SQLite for persistent storage with CRUD functionality for tasks.
- Designed an intuitive UI supporting task categorization and prioritization features.